SMART INDIA HACKATHON · SIH 26262

AICTE Approval & Compliance Orchestration Platform

Complete REST API Specification & Developer Reference

AICTE Approval & Compliance Orchestration Platform (SIH 26262)

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Specification Standard: OpenAPI 3.0 / REST / JSON

Base URL (Local): `http://localhost:8080/api/v1`

Swagger UI: `http://localhost:8080/swagger-ui.html`

OpenAPI JSON: `http://localhost:8080/v3/api-docs`

Protocol: HTTP/1.1 (HTTPS in Production)

Authentication Scheme: HTTP Bearer Token (`Authorization: Bearer <TOKEN>`)

Error Standard: RFC 7807 Problem Details (`application/problem+json` & `application/json`)

Backend Framework: Spring Boot 3.4.3 + Spring Security 6.4.3 + PostgreSQL 18.6

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1. Architectural Principles & Design Standards

The platform's API surface adheres to strict enterprise REST guidelines:

- **Stateless Communication**: No server-side HTTP session state. Every request carries an `Authorization: Bearer` header.
- **Resource-Oriented URIs**: Plural nouns for collections (`/institutions` , `/applications`) and subordinate actions for operations (`/applications/{id}/actions`).
- **Standard HTTP Verbs**:
- `GET` : Safe, idempotent read operations.
- `POST` : Non-idempotent creation or consequential state transitions.
- **HTTP Status Codes**:
- `200 OK` : Successful synchronous retrieval or state mutation.
- `201 Created` : Successful creation of a persistent entity, accompanied by a `Location` header pointing to the canonical resource URI.
- `400 Bad Request` : Validation failure, parameter type mismatch, or invalid state machine precondition.
- `401 Unauthorized` : Missing, unparseable, or invalid Bearer token.
- `404 Not Found` : Referenced entity ID or application code does not exist.
- `405 Method Not Allowed` : HTTP method unsupported for target URI.
- `409 Conflict` : Unique constraint violation or JPA optimistic locking collision.
- `415 Unsupported Media Type` : Request `Content-Type` is not `application/json` .
- `500 Internal Server Error` : Unhandled server exception.

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2. Authentication & Development Token Bridge

Overview

To facilitate rapid multi-role testing during hackathon evaluations without external identity provider dependencies, the backend implements a server-managed development authentication bridge in `DevTokenService.java` and `DevBearerAuthenticationFilter.java`.

Supported Demonstration Roles

The bridge recognizes the following canonical roles:

- `Scrutiny Officer` (Principal: `USR-SO-01`, B. Singh)
- `Institution Representative` (Principal: `USR-INST-01`, Dr. Ramesh)
- `Regional Officer` (Principal: `USR-RO-01`, Dr. A. Sharma)
- `EVC Coordinator` (Principal: `USR-EVC-01`, C. Verma)
- `Compliance Reviewer` (Principal: `USR-COMP-01`, P. Mehta)
- `Senior Management` (Principal: `USR-SM-01`, Prof. M. K. Sridhar)
- `Administrator` (Principal: `USR-ADM-01`, System Administrator)

Token Issuance Flow

1. Client sends `POST /api/v1/auth/token` with `{"role": "Scrutiny Officer"}`.
2. Server validates the role, resolves the user profile, issues a signed token `dev-token-USR-SO-01-<uuid>`, and caches it in memory.
3. Client stores this token in browser `localStorage` under `aicte_auth_token`.
4. Client passes `Authorization: Bearer <TOKEN>` in subsequent requests.
5. `DevBearerAuthenticationFilter` intercepts the request, verifies the token against the registry, and populates the `SecurityContextHolder` with `ROLE_SCRUTINY_OFFICER`.

3. Error Response Format (RFC 7807)

All non-2xx responses conform to the **RFC 7807 Problem Details for HTTP APIs** specification:

JSON

```
{
  "type": "https://api.aicte.gov.in/errors/validation-failed",
  "title": "Validation Failed",
  "status": 400,
  "detail": "Request validation failed. Inspect field errors.",
  "instance": "/api/v1/applications",
  "timestamp": "2026-10-23T23:08:50.123Z",
  "invalidFields": {
    "applicationReference": "must not be blank",
    "institutionId": "must not be null"
  }
}
```

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4. Pagination, Sorting & Filtering Protocol

Endpoints returning collections (`GET /api/v1/institutions` , `GET /api/v1/applications`) support standard Spring Data pagination parameters:

Query Parameters

- `page` (integer, default `0`): 0-indexed page number.
- `size` (integer, default `20` , maximum `100`): Maximum items returned per page.

Envelope Structure (`PageResponse<T>`)

JSON

```
{
  "items": [ ... ],
  "pageNumber": 0,
  "pageSize": 20,
  "totalElements": 42,
  "totalPages": 3,
  "isFirst": true,
  "isLast": false,
  "hasNext": true,
  "hasPrevious": false
}
```

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5. Complete Endpoint Reference

5.1 System Health

`GET /api/v1/health`

- **Purpose**: System liveness and diagnostic probe.

- **Authentication**: None (Public).
- **Responses**:
- `200 OK`: System healthy.
- **Example cURL**:

```bash
curl -s http://localhost:8080/api/v1/health
```

- **Example Response**:

```json
{
  "status": "UP",
  "application": "aicte-compliance-platform",
  "timestamp": "2026-10-23T23:08:50.000Z"
}
```

5.2 Issue Development Token

`POST /api/v1/auth/token`

- **Purpose**: Issue an authentic Bearer token for a demo role.
- **Authentication**: None (Public Dev Bridge).
- **Headers**: `Content-Type: application/json`
- **Request Body**:

```json
{
  "role": "Scrutiny Officer"
}
```

- **Responses**:
- `200 OK`: Token generated.
- **Example cURL**:

```bash
curl -s -X POST http://localhost:8080/api/v1/auth/token   -H "Content-Type: application/js
```

- **Example Response**:

JSON


```
{
  "accessToken": "dev-token-USR-SO-01-f5479a8a",
  "tokenType": "Bearer",
  "expiresIn": 86400,
  "user": {
    "id": "USR-SO-01",
    "name": "B. Singh",
    "email": "scrutiny.singh@aicte-india.gov.in",
    "role": "Scrutiny Officer",
    "region": "Northern Region",
    "designation": "Senior Scrutiny Officer"
  }
}
```


5.3 Get Current User Profile

`GET /api/v1/auth/me`

- **Purpose**: Verify token validity and fetch caller profile.
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Responses**:
- `200 OK`: Authenticated profile returned.
- `401 Unauthorized`: Token missing or invalid.
- **Example cURL**:

BASH

```
curl -s http://localhost:8080/api/v1/auth/me   -H "Authorization: Bearer dev-token-USR-SO-
```

5.4 List Institutions

`GET /api/v1/institutions`

- **Purpose**: Retrieve paginated list of canonical institutions.
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Query Parameters**: `page` (int, default 0), `size` (int, default 20)
- **Responses**:
- `200 OK`: `PageResponse<InstitutionResponse>`
- `401 Unauthorized`: Missing token.
- **Example cURL**:

BASH

```
curl -s "http://localhost:8080/api/v1/institutions?page=0&size=10"   -H "Authorization: Be
```

5.5 Register Institution

`POST /api/v1/institutions`

- **Purpose**: Register a new canonical institution in the directory.
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Headers**: `Content-Type: application/json`
- **Request Body**:

```json
{
  "institutionalIdentifier": "INST-1099",
  "name": "National Technical Institute",
  "region": "Northern Region"
}
```

- **Responses**:
- `201 Created`: Institution created (Returns `Location` header).
- `400 Bad Request`: Field validation failure.
- `409 Conflict`: Duplicate `institutionalIdentifier`.
- **Example cURL**:

```bash
curl -s -i -X POST http://localhost:8080/api/v1/institutions   -H "Authorization: Bearer $
```

5.6 Get Institution by ID

`GET /api/v1/institutions/{id}`

- **Purpose**: Retrieve institution details by numeric primary key.
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Path Variable**: `id` (Long primary key)
- **Responses**:
- `200 OK`: `InstitutionResponse`
- `404 Not Found`: Institution not found.
- **Example cURL**:

```bash
curl -s http://localhost:8080/api/v1/institutions/1   -H "Authorization: Bearer $TOKEN"
```

5.7 List Applications (With Search & Filtering)

`GET /api/v1/applications`

- **Purpose**: Retrieve paginated list of applications with relational filtering.
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Query Parameters**:
- `institutionId` (Long): Filter by institution primary key.
- `status` (String): Filter by application status (`In progress` , `Returned` , `Due soon`).
- `academicYear` (String): Filter by academic cycle (`2026-2027`).
- `region` (String): Filter by institution region (`Northern Region`).
- `stage` (String): Filter by stage (`Scrutiny` , `Re-scrutiny` , `EVC scheduling`).
- `q` (String): Multi-field case-insensitive search.
- `page` (Integer): Page number (default 0).
- `size` (Integer): Page size (default 20).
- **Responses**:
- `200 OK` : `PageResponse<ApplicationResponse>`
- **Example cURL**:

BASH

```
curl -s "http://localhost:8080/api/v1/applications?region=Northern+Region&stage=Scrutiny"
```

5.8 Register Application

`POST /api/v1/applications`

- **Purpose**: Register a new application linked to a canonical institution.
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Request Body**:

JSON

```
{
  "institutionId": 1,
  "applicationReference": "APP-2026-9001",
  "applicationType": "Extension of Approval / Compliance Review",
  "academicYear": "2026-2027"
}
```

- **Responses**:
- `201 Created` : Application registered with status `In progress` , stage `Scrutiny` .
- `400 Bad Request` : Field validation failure.
- `404 Not Found` : Referenced `institutionId` does not exist.
- `409 Conflict` : Duplicate `applicationReference` .
- **Example cURL**:

BASH

```
curl -s -i -X POST http://localhost:8080/api/v1/applications   -H "Authorization: Bearer $
```

5.9 Get Application Details & Timeline

`GET /api/v1/applications/{id}`

- **Purpose**: Fetch application dossier including linked institution and chronological workflow transitions.
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Path Variable**: `id` (Supports numeric ID `13` or reference string `APP-2026-1042`).
- **Responses**:
- `200 OK`: Full `ApplicationResponse` with `timeline`.
- `404 Not Found`: Application reference not found.
- **Example cURL**:

BASH

```
curl -s http://localhost:8080/api/v1/applications/APP-2026-1042   -H "Authorization: Beare
```

5.10 Execute Workflow Action

`POST /api/v1/applications/{id}/actions`

- **Purpose**: Execute a state transition (`RETURN_FOR_CORRECTION`, `SUBMIT_CORRECTIONS`, `COMPLETE_SCRUTINY`).
- **Authentication**: `Authorization: Bearer <TOKEN>`
- **Path Variable**: `id` (Supports numeric ID or reference string).
- **Headers**: `Content-Type: application/json`
- **Request Body (`WorkflowActionRequest`)**:

JSON

```
{
  "actionType": "RETURN_FOR_CORRECTION",
  "actorName": "B. Singh",
  "actorRole": "Scrutiny Officer",
  "reason": "Fire safety certificate expired.",
  "newDeficiencies": [
    "Fire Safety NOC issued by Delhi Fire Service expired on 2026-09-30"
  ],
  "observations": "Land and building verified. Safety compliance non-compliant."
}
```

- **Responses**:
- `200 OK`: Action executed atomically; returns updated `ApplicationResponse`.
- `400 Bad Request`: Missing reason, invalid action, or precondition violation.

- `404 Not Found`: Application not found.
- `409 Conflict`: JPA optimistic locking collision.
- **Example cURL**:

```bash
curl -s -X POST http://localhost:8080/api/v1/applications/APP-2026-1042/actions   -H "Auth
```

---

6. Workflow State Machine Action Payloads

Action 1: `RETURN_FOR_CORRECTION`

- **Prerequisites**: Application stage must be `Scrutiny` or `Re-scrutiny`.
- **Target State**: Stage becomes `Re-scrutiny`, status becomes `Returned`, owner becomes `Institution Representative`.
- **Payload Schema**:

```json
{
  "actionType": "RETURN_FOR_CORRECTION",
  "actorName": "string (optional, defaults to authenticated user)",
  "actorRole": "string (optional)",
  "reason": "string (MANDATORY, non-blank)",
  "newDeficiencies": ["string array of specific objections"],
  "observations": "string (optional review remarks)"
}
```

Action 2: `SUBMIT_CORRECTIONS`

- **Prerequisites**: Application status must be `Returned`.
- **Target State**: Stage remains `Re-scrutiny`, status becomes `In progress`, owner becomes `Scrutiny Officer`.
- **Payload Schema**:

```json
{
  "actionType": "SUBMIT_CORRECTIONS",
  "actorName": "string (optional, e.g. Dr. Ramesh)",
  "actorRole": "Institution Representative",
  "reason": "string (MANDATORY, explanation of rectifications made)"
}
```

Action 3: `COMPLETE_SCRUTINY`

- **Prerequisites**: Application stage must be `Scrutiny` or `Re-scrutiny`. Open deficiencies must be resolved.

- **Target State**: Stage advances to `nextStage` (default: `EVC scheduling`), status becomes `Due soon`, owner becomes `EVC Coordinator`.
- **Payload Schema**:

JSON

```
{
  "actionType": "COMPLETE_SCRUTINY",
  "actorName": "B. Singh",
  "actorRole": "Scrutiny Officer",
  "nextStage": "EVC scheduling",
  "observations": "string (optional final endorsement)"
}
```

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7. Automated Test Evidence Matrix

Test Method
`HealthEndpointTest.shouldReturnHealthStatusOk`
`HealthEndpointTest.shouldRejectUnauthenticatedAccess`
`WorkflowAndSecurityIntegrationTest.shouldAuthenticateWithDevBearerToken`
`WorkflowAndSecurityIntegrationTest.shouldReturn400ForMalformedJson`
`WorkflowAndSecurityIntegrationTest.shouldReturn400ForParameterTypeMismatch`
`WorkflowAndSecurityIntegrationTest.shouldReturn405ForUnsupportedMethod`
`WorkflowAndSecurityIntegrationTest.shouldReturn415ForUnsupportedMediaType`
`WorkflowAndSecurityIntegrationTest.shouldHandleSpecialCharactersInAuditMetadataSafely`
`WorkflowAndSecurityIntegrationTest.shouldExecuteFullScrutinyLifecycleTransitions`

Test Method
`InstitutionAndApplicationApiIntegrationTest.shouldCreateAndRetrieveInstitution`
`InstitutionAndApplicationApiIntegrationTest.shouldRejectDuplicateInstitutionalIdentifier`
`InstitutionAndApplicationApiIntegrationTest.shouldCreateAndRetrieveApplicationLinkedToInstitu`
`InstitutionAndApplicationApiIntegrationTest.shouldRejectDuplicateApplicationReference`
`InstitutionAndApplicationApiIntegrationTest.shouldPaginateAndFilterApplications`

End of SIH 26262 API Documentation.